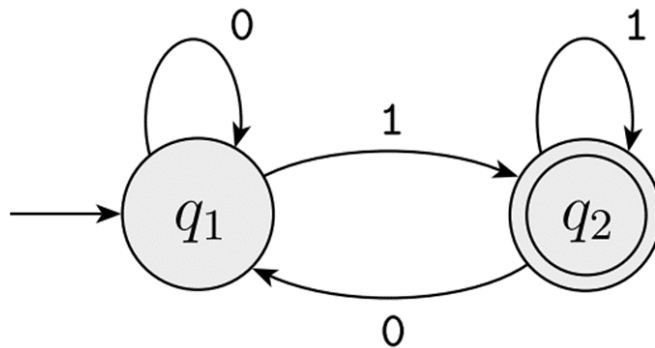


Example Problems of Regular Language to DFA

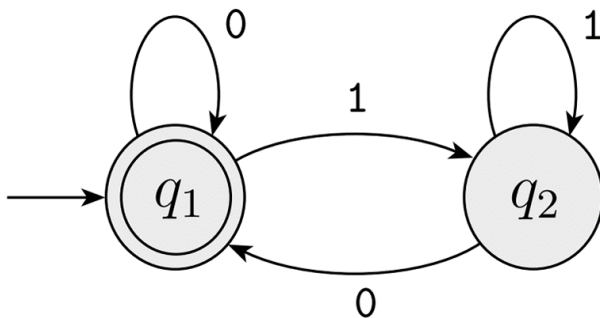
1. $L_1 = \{w \in \{0, 1\}^* \mid w \text{ ends with a } 1\}$

Notice here, this language set contains strings such as 1, 01, 001, 101, 111, 0101, ...

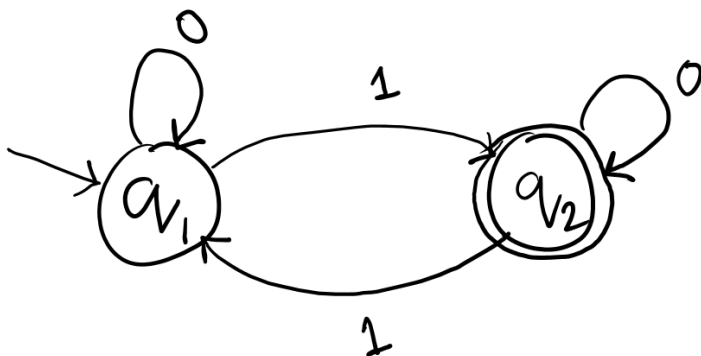


2. $L_2 = \{w \in \{0, 1\}^* \mid w \text{ does not end with a } 1\}$

Notice the language of L_1 and L_2 . L_2 is actually complement set of L_1 . Now Notice the DFA's of both languages; they are the same DFA's except the accept state. If you have designed a DFA for a language, you can easily create a DFA for its complement set. The states that were previously accept states, will not be accept states anymore and the states that were not accept state before will become accept states now.

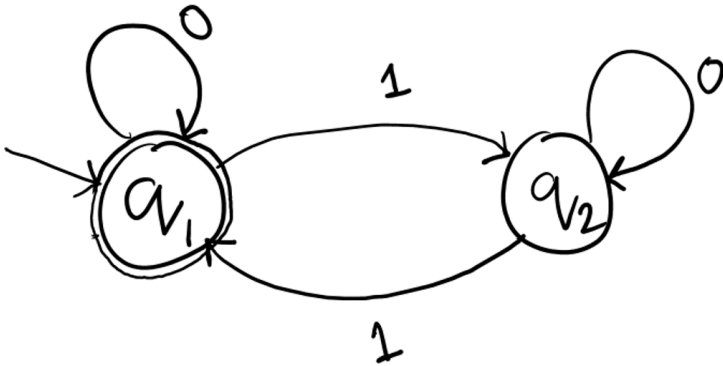


3. $L_3 = \{w \in \{0, 1\}^* \mid w \text{ has odd number of } 1\text{'s}\}$

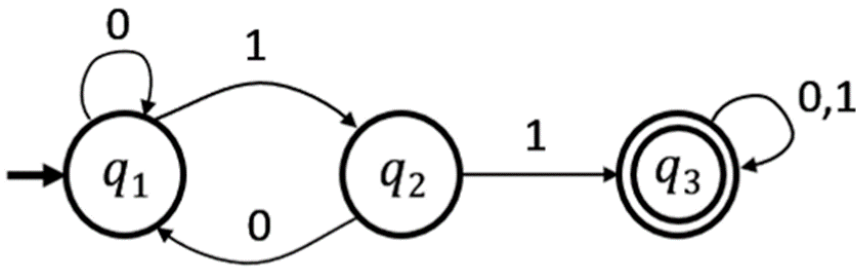


Example Problems of Regular Language to DFA

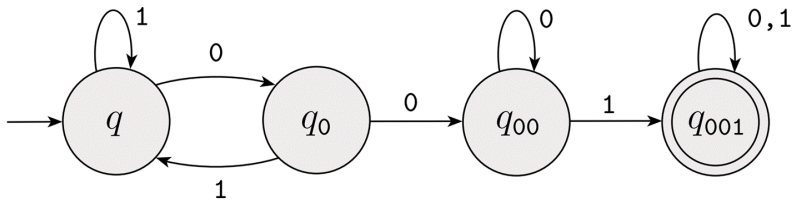
4. $L_4 = \{w \in \{0, 1\}^* \mid w \text{ has even number of 1's}\}$



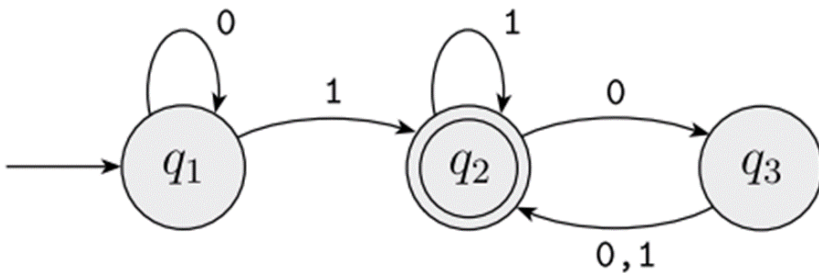
5. $L_5 = \{w \in \{0, 1\}^* \mid w \text{ has substring 11}\}$



6. $L_6 = \{w \in \{0, 1\}^* \mid w \text{ has substring 001}\}$

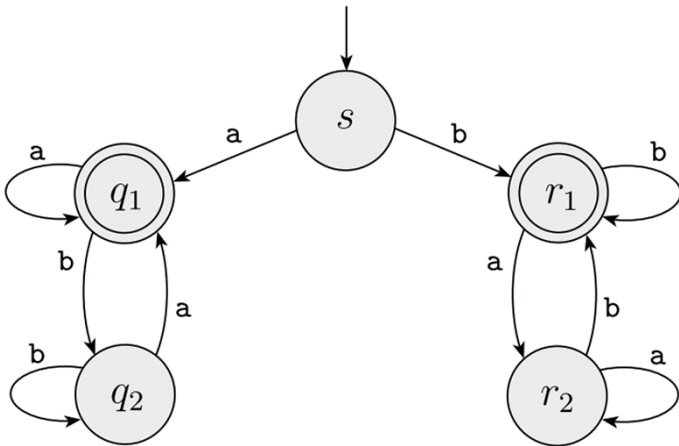


7. $L_7 = \{w \in \{0, 1\}^* \mid \text{contains at least one 1 and even number of 0's follow the last 1}\}$



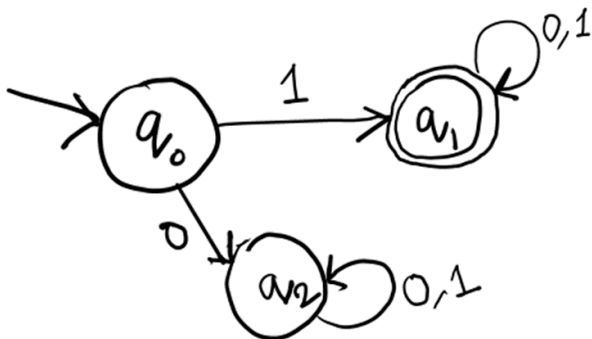
Example Problems of Regular Language to DFA

8. $L_8 = \{w \in \{a, b\}^* \mid w \text{ starts and ends with } a \text{ or } w \text{ starts and ends with } b\}$



9. $L_9 = \{w \in \{0, 1\}^* \mid w \text{ starts with } 1\}$

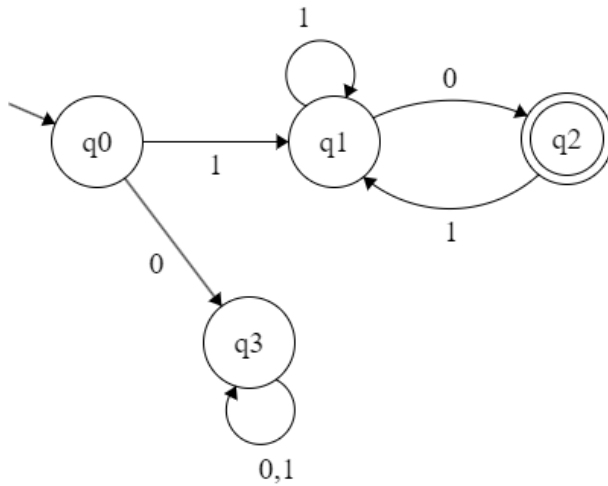
Notice that, we are transitioning to q_3 when we are getting 0 at the beginning of the string. After getting 0 at the beginning of the string there is no way the sentence is going to be accepted. So, q_3 is acting like a Trap state or Dead state here. After transitioning to these kinds of states there is no way to go to the accept state.



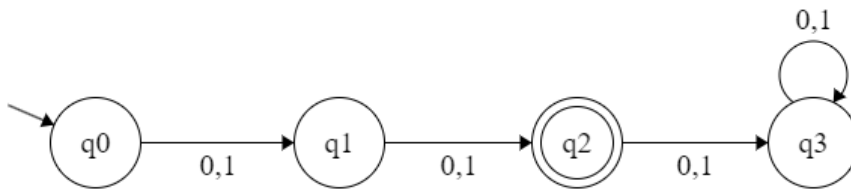
Example Problems of Regular Language to DFA

10. $L_{10} = \{w \in \{0, 1\}^* \mid w \text{ starts with } 1 \text{ and ends with } 0\}$

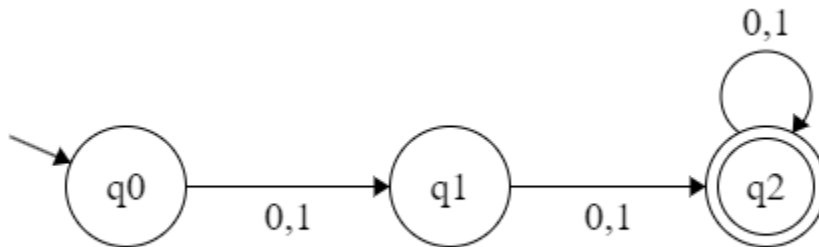
In this diagram in q_2 , there will be a loop with 0 transition.



11. $L_{11} = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is two}\}$

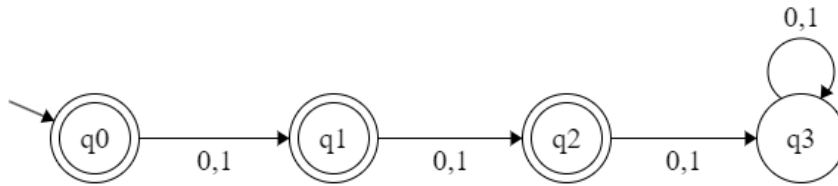


12. $L_{12} = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is at least two}\}$

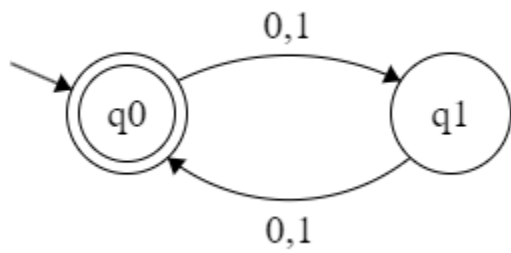


Example Problems of Regular Language to DFA

13. $L_{13} = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is at most two}\}$



14. $L_{14} = \{w \in \{0, 1\}^* \mid w \text{ has even length}\}$



15. $L_{15} = \{w \in \{0, 1\}^* \mid w \text{ has odd length}\}$

